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FACULDADE DE CIÊNCIAS
UNIVERSIDADE DO PORTO

Faculty of Sciences of the University of Porto

Master's Degree in Information Security

Systems & Data Security

Public Ledger for Auctions over Kademlia

Marco Rui Fernandes

May, 2023

Abstract

In this report, we briefly describe the application we developed and discuss the decisions we made.

Contents

List of Figures

Acronyms

P2P Peer-To-Peer

PoS Proof of Stake

PoW Proof of Work

1 Introduction

Kademlia [1]

This report is structured as follows: Finally, §7 concludes the report.

2 Problem & Proposed Solution

Similar to Peermart [2].

Our implementation is divided in three parts:

– **P2P network:**

– **Public Ledger:** The public ledger should be able to store auction transactions.

– **Auction Service:**

3 Kademlia Network

3.1 Sybil & Eclipse Attacks

[3].

4 Public Ledger

Regarding PoS, we implemented

5 Reputation System

6 Auction System

7 Conclusion & Future Work

Unfortunately, due to timing constraints/time limitations, we were not able to test our application using Google Cloud's virtual machines. Nevertheless,

... using an identity-based approach, such as the one discussed in [4].

References

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[2] David Hausheer and Burkhard Stiller. Peermart: The technology for a distributed auction-based market for peer-to-peer services. In *IEEE International Conference on Communications, 2005. ICC 2005. 2005*, volume 3, pages 1583–1587. IEEE, 2005.

[3] Riccardo Pecori. S-Kademlia: A trust and reputation method to mitigate a Sybil attack in Kademlia. *Computer Networks*, 94:205–218, 2016.

[4] Luca Maria Aiello, Marco Milanesio, Giancarlo Ruffo, and Rossano Schifanella. Tempering Kademlia with a Robust Identity Based System. In *2008 Eighth International Conference on Peer-to-Peer Computing*, pages 30–39, 2008.